Gwyneth Margaux Tangog (Gwyn)

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EDUCATION

Massachusetts Institute of Technology — Cambridge, MA
Master of Engineering in Computer Science — Expected Graduation: May 2027, GPA: 5.0/5.0
Bachelor of Science in Computer Science and Engineering & Mathematics — Graduated: May 2026, GPA: 4.8/5.0
Relevant Coursework: Software Construction, Computer Systems Engineering, Design and Analysis of Algorithms, Sublinear Time Algorithms, Machine Learning, Advances in Computer Vision, AI in Medicine, Probability and Random Variables

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PROFESSIONAL EXPERIENCE

MIT Materials Systems Laboratory, *Graduate Research Assistant* — June 2026 - current

- Architected and deployed a full-stack dashboard (Python backend, JavaScript frontend) analyzing material substitution patterns across 4 global regions; processed 20+ synthetic interview datasets to identify regional utility functions
- Designed schema and data pipeline to ingest, normalize, and visualize multi-regional qualitative and quantitative data; enabled real-time filtering and comparative analysis for research team (8+ users).

MIT Theory of Computation Group, *Undergraduate Researcher* — Jan - May 2026

- Proved novel hardness results using complexity reductions and implemented polynomial-time algorithms for Unlimited Solo Chess; manuscript **submitted to Fun with Algorithms, currently under revision**.
- Proposed a novel well-defined Weight Balancing problem and proved NP-hardness.
- Derived lower bounds using the Exponential Time Hypothesis for the Polyomino Tiling problem.

Recidiviz, *Implementation Engineer Intern* — June - Aug 2024

- Implemented and shipped data ingestion pipelines and views for criminal justice datasets on the Recidiviz platform, serving 19 states, including writing tests in JavaScript and participating in code reviews.
- Designed and iterated on SQL schema changes to track program completions.

MIT Concrete Sustainability Hub, *Undergraduate Researcher* — Sept 2023 - June 2024

- Built end-to-end Python data pipelines for 2 research projects on concrete carbon uptake and concrete delivery workforce retention; published a brief and presented research posters to **30+ stakeholders** at Technical Advisory Group meetings.
- Modeled lithium price elasticity using bottom-up demand estimation; conducted literature review and sensitivity analysis; published in Resources, Conservation & Recycling (2024).

The Engine Room, *Data and Tech Intern* — Jan - Feb 2024

- Built Python automation scripts for analysis of the Light Touch Support program reporting data; created visualizations and summary metrics to identify key program outcomes and areas for improvement, reducing monthly **reporting analysis time from 4+ hours to 30 minutes**; documented scripts for team reuse.
- Designed and deployed interactive data analysis dashboard enabling non-technical program staff to easily access results (used by 5+ stakeholders).

LEADERSHIP & MENTORSHIP

MIT EECS Department, *Teaching Assistant, Math for Computer Science* — Sept 2024 - May 2026

- Led weekly recitation sections of 15+ students and office hour blocks with 20+ students across **4 semesters**; received a **7.0/7.0 final course rating**.
- Initiated, wrote material for, and led well-attended **(100+ students)** classwide review sessions which covered proof methods, algorithmic analysis, graph theory, number theory, combinatorics, and probability.
- Collaborated with course instructors on grading, writing problems, and student accommodation strategies.

MIT Office of Minority Education, *Residential Facilitator & Teaching Assistant, Calculus II* — June 2025 - Aug 2025

- Led weekly recitation sections of 15+ students and office hour blocks for MIT's Interphase EDGE Program, which assists incoming freshmen from underserved communities in acclimating to the rigor at MIT.
- Prepared and streamlined problems, content, and solutions for recitation sessions, reducing weekly preparation time for future TAs by 3+ hours.
- Mentored a cluster of 4 incoming freshmen; organized 2 outings for the cluster with other cluster leaders.

AWARDS & HONORS

Outstanding UROP (Undergraduate Research Opportunity) Award, 2024 | MIT Eta Kappa Nu (HKN) Honors Society | China Girls Math Olympiad - Silver Medal, 2019 | Philippine Math Olympiad - National Finalist, 2019

SKILLS

Data Analysis Languages: Python (expert), SQL (expert), PostgreSQL (expert)

Data Analysis and Machine Learning Tools: pandas, seaborn, matplotlib, PyTorch, scikit-learn

Software Engineering Languages: TypeScript (proficient), JavaScript (proficient), C (proficient), Assembly (proficient)

Infrastructure & Other Tools: Git, GitHub, Astro, Github Pages, Render, Testing (pytest), LaTeX